
Polymorphic Malware Behavior Simulator

What It Does and Why It Matters for Your Demo

A guide for Sales Engineers | Simple language | Step-by-step explanations

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What Is This Tool?

This tool is a **demo simulator**. It pretends to be malware (bad software) so you can show customers how Trend Micro catches real threats. It does **NOT** cause real damage. Everything it does is in a safe test environment.

***TIP:** Think of it like a **fire drill**. A fire drill is not a real fire, but it tests if the alarm system works. This tool is a 'cyberattack drill' — it tests if Trend Micro Vision One detects and alerts on real attack behaviors.*

Key Facts

- Safe to run on a dedicated test machine
- Does **not** spread to other computers
- All files it creates can be removed with one cleanup command
- Designed to **trigger** Trend Micro alerts — not to avoid them
- Every action is logged so you can explain exactly what happened

Why We Use This Tool

Many customers ask: "How do I know Trend Micro actually works against real malware?" Showing old screenshots is not convincing. They want to see it **live**.

This tool creates real attack behaviors — the same techniques that real hackers use. When you run it on a test machine with Trend Micro installed, Vision One will:

1. Detect each attack technique in real time
2. Create alerts (called **Workbenches**)
3. Show the full story of what happened and when

"The customer sees Trend Micro catching a real attack, live, in front of them."

The 6 Attack Modules

This tool runs 6 different attack behaviors. Each one represents a real technique that cybercriminals use. They run one after another automatically.

1	EICAR File Drop	Drops a standard test malware file	Malware Detection
2	Polymorphic Mutation	Copies itself with a different fingerprint	Suspicious Behavior (T1027)
3	Persistence	Installs 3 ways to survive a computer restart	Suspicious Behavior (T1547, T1053)
4	Process Injection	Hides attack code inside a trusted program	Suspicious Behavior (T1055)
5	C2 Beacon	Sends regular signals to attacker servers	Network Correlation (T1071)
6	Ransomware Simulation	Encrypts 75 files to trigger ransomware alerts	Ransomware Indicators (T1486)

MODULE 1

EICAR File Drop

EICAR is a special test file used worldwide by the security industry. It is not real malware. Every major antivirus product recognizes this file and will create an alert when it sees it. It is completely safe to use.

What the Tool Does:

- The tool builds the EICAR test file at the moment it runs (not before)
- It saves the file to the Windows Temp folder
- Trend Micro sees the file immediately and creates a malware alert

Why This Is Important for Your Demo:

- This is the **fastest and most reliable** alert to generate
- The customer sees Trend Micro react in seconds — good opening moment
- Proves that real-time file scanning is working correctly

What Vision One Shows:

- Alert type: **Malware Detection** (severity: High)
- File path: C:\Users\...\Temp\eicar_test_XXXXXXXXX.com
- Detection name: EICAR_TEST_FILE
- Action taken: File blocked and quarantined
- Time to detect: Usually under 10 seconds

"This is like showing you a known criminal's face. Trend Micro recognized it immediately."

MODULE 2

Polymorphic Mutation Chain

Polymorphic means 'changes shape.' Real malware often changes itself so that security tools cannot recognize it by its fingerprint (called a **hash**). Every time the malware copies itself, it looks different — but it behaves the same way.

What the Tool Does:

- The original file creates a copy of itself with a **different fingerprint**
- That copy creates another copy — also with a different fingerprint
- This creates a chain of 3 copies, each with a unique fingerprint
- Each copy automatically runs and creates the next generation

Why This Is Important for Your Demo:

- Old security tools use 'hash matching' — they only catch malware they have seen before
- Trend Micro uses **behavior analysis** — it detects what the malware DOES, not just what it looks like
- This proves Trend Micro can catch **brand new malware** nobody has seen before

What Vision One Shows:

- Multiple new .exe files appearing in the Temp folder
- Each file has a completely different SHA-256 fingerprint
- Trend Micro links all copies together as ONE attack story in the Workbench
- MITRE ATT&CK; technique: T1027 (Obfuscated Files or Information)
- Process chain visible: original → gen1 → gen2 → gen3

"This is like a criminal who changes their clothes and disguise each day. Old security only knows one face. Trend Micro watches what the person DOES — and catches them every time."

MODULE 3

Persistence

When attackers get inside a computer, they want to **STAY there** — even if the computer restarts. Persistence means setting up ways to automatically restart the attack software after a reboot.

What the Tool Does:

- **Registry Run Key:** Adds an entry to Windows Registry so the program starts at every login
- **Scheduled Task:** Creates a Windows task named 'WinSvcHost32' that runs at login
- **Startup Folder:** Copies itself to the Windows Startup folder

Why This Is Important for Your Demo:

- Real attackers always install persistence — they want to keep access
- Shows Trend Micro can detect **3 different persistence methods** at once
- Customers who have been attacked before will recognize this as a real threat

What Vision One Shows:

- Alert type: **Suspicious Behavior** — Persistence
- MITRE ATT&CK:: T1547.001 (Registry Run Key) and T1053.005 (Scheduled Task)
- Details show the exact registry path and task name
- Timeline shows all 3 persistence events with timestamps

"The attacker installed three different 'back doors' so they can always come back in. Trend Micro caught all three."

MODULE 4 Process Injection

This is an advanced attack technique. The malware hides its code **INSIDE** another program that is already running — like Notepad (a trusted Windows program). This is very hard to detect because the attack code runs inside a program that looks safe.

What the Tool Does:

- Opens Notepad (a safe, trusted Windows program)
- Writes a tiny piece of code inside Notepad's memory space
- Makes Notepad run that code
- **NOTE:** The code is completely harmless — it runs for less than 1 second and does nothing
- The **technique** is what Trend Micro detects, not the payload

Why This Is Important for Your Demo:

- Traditional security tools only watch files on disk — they miss in-memory attacks
- Trend Micro's behavioral engine watches **what programs do in memory**
- This is the same technique used by ransomware groups like LockBit and REvil
- **Requires running as Administrator** — tell customer before demo

What Vision One Shows:

- Alert type: **Suspicious Behavior** — Process Injection
- MITRE ATT&CK:: T1055.002 (Portable Executable Injection)
- Process tree shows: WinSvcHost32.exe → notepad.exe (injection detected)
- API call chain logged: OpenProcess → VirtualAllocEx → WriteProcessMemory → CreateRemoteThread

"Imagine a thief wearing a police uniform. Old security sees only the uniform — a trusted program. Trend Micro watches what the program DOES inside, not just who it appears to be."

MODULE 5

C2 Beacon

C2 stands for Command and Control. Real malware regularly contacts the attacker's server — like a soldier checking in with headquarters. This regular contact signal is called a **beacon**.

What the Tool Does:

- Every 30 seconds, the tool sends HTTP requests to 3 different targets:
- 1. Trend Micro's own WRS test URL (guaranteed to trigger a network alert)
- 2. A known test host: `wrs49.winshipway.com`
- 3. A configurable custom target (default: `puginarug.com`)
- Uses suspicious browser identification strings that look like old malware
- Adds custom headers that look like real malware communication

Why This Is Important for Your Demo:

- Real malware uses multiple backup servers — if one is blocked, it tries another
- The **regular pattern** of connections is what identifies C2 beaconing
- Demonstrates that Trend Micro catches network threats, not just file threats

What Vision One Shows:

- Alert type: **Network Correlation** / C2 Detection
- MITRE ATT&CK;: T1071.001 (Web Protocols)
- Shows source IP (your test machine) and all beacon destinations
- Web Reputation (WRS) score shown for each destination
- Connection frequency graph proves it is not random — it is beaconing

"This is like a spy who sends a message to headquarters every 30 minutes, saying 'I am still here, waiting for orders.' Trend Micro notices the pattern and blocks the communication."

MODULE 6

Ransomware Simulation

Ransomware is malware that encrypts (scrambles) your files so you cannot open them. The attacker then demands payment to unscramble the files. This is one of the most damaging cyberattacks for businesses today.

What the Tool Does:

- Creates 75 test files in a safe temporary folder (never touches real user files)
- Files have realistic names: Document_001.docx, Photo_023.jpg, Report_045.xlsx
- Reads each file and scrambles (encrypts) the content
- Renames each file with a new extension: **.polyenc**
- Creates a 'ransom note' file: README_DECRYPT.txt
- **NOTE:** Recovery is easy — the encryption key is very simple (for demo only)

Why This Is Important for Your Demo:

- The **speed and volume** of file operations is what triggers ransomware detection
- Even if the ransomware were completely new (no signature), Vision One detects the behavior
- Customers always react strongly to seeing ransomware detected — very impactful demo moment

What Vision One Shows:

- Alert type: **Ransomware Indicators**
- MITRE ATT&CK: T1486 (Data Encrypted for Impact), T1083 (File Discovery)
- Count of files affected shown (75 files)
- New extension (.polyenc) flagged as suspicious
- Location: C:\Users\...\Temp\Documents_Backup_XXXXXXX\
- Ransom note file (README_DECRYPT.txt) detected as an indicator

"Real ransomware would scramble thousands of important documents in minutes. Trend Micro detected the pattern after just a few files. In a real scenario, it would stop the ransomware before most files are damaged."

How to Run the Demo Tool

Requirements

- Windows 10 or Windows 11 test machine (dedicated lab VM recommended)
- Trend Micro SEPA or Apex One agent installed and connected to Vision One
- XDR / Endpoint Sensor and Behavior Monitoring enabled in the agent policy
- Run as **Administrator** for full results (required for Process Injection module)

Running the EXE (Recommended for Demo)

1. Copy **polymorphic-demo.exe** to the Windows test machine
2. Right-click the file → select **Run as Administrator**
3. Watch the console — each module shows its progress as it runs
4. Switch to Vision One console and wait **2–5 minutes** for alerts to appear

Command Options

<code>polymorphic-demo.exe</code>	Run all 6 modules (recommended)
<code>polymorphic-demo.exe -no-inject</code>	Skip process injection (use if not running as Administrator)
<code>polymorphic-demo.exe -interval 15</code>	Beacon every 15 seconds instead of 30 (faster demo)
<code>polymorphic-demo.exe -cleanup</code>	Remove ALL artifacts after the demo
<code>polymorphic-demo.ps1 -Cleanup</code>	PowerShell version cleanup

■ **IMPORTANT:** Always run **polymorphic-demo.exe -cleanup** after every demo. This removes all registry entries, scheduled tasks, temp files, and stops the beacon. Vision One will keep the historical alerts — you can still use them for conversation.

Where Files Are Generated

All artifacts are written inside the Windows **%TEMP%** folder or **%APPDATA%**. No files are created outside of these locations. The exact path is always printed in the console when each module runs.

Polymorphic Mutation Files

Each generation is written to **%TEMP%** as a new EXE and immediately executed. Generations appear 2 seconds apart.

gen1	%TEMP%\WinSvcHost_gen1_<random>.exe	Original EXE
gen2	%TEMP%\WinSvcHost_gen2_<random>.exe	gen1
gen3	%TEMP%\WinSvcHost_gen3_<random>.exe	gen2 (final)

TIP: Each file has a different SHA-256 hash but identical behavior. To find them quickly: open File Explorer → paste **%TEMP%** in the address bar → look for WinSvcHost_gen*.exe files.

Ransomware Simulation Files (.polyenc)

Created inside a randomly-named subfolder in **%TEMP%**. The exact folder name is printed in the console when the module runs.

%TEMP%\Documents_Backup_<random>

Document_001.polyenc ... Document_075.polyenc ← 75 XOR-encrypted test files (key: 0x42)

README_DECRYPT.txt ← Simulated ransom note — triggers ransomware heuristics

All Artifact Locations at a Glance

EICAR test file	%TEMP%\eicar_test_<random>.com
Mutation EXEs (gen1–3)	%TEMP%\WinSvcHost_gen<N>_<random>.exe
.polyenc encrypted files	%TEMP%\Documents_Backup_<random>*.polyenc
Persistent EXE copy	%APPDATA%\Microsoft\Windows\WinSvcHost32.exe
Startup folder copy	%APPDATA%\Microsoft\Windows\Start Menu\Programs\Startup\WinSvcHost32.exe
Registry run key	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\WinSvcHost32

Scheduled task	Task Scheduler → WinSvcHost32 (ONLOGON)
Demo log file	%TEMP%\polymorphic_demo_log.txt

■ **IMPORTANT:** Run **polymorphic-demo.exe -cleanup** to remove every artifact in this table. To verify manually: open %TEMP% in File Explorer and confirm no WinSvcHost_* or Documents_Backup_* folders remain.

Demo Package Contents

polymorphic-demo.exe	Windows x64 compiled executable — run on test machine
polymorphic-demo.ps1	PowerShell version — no build required, copy and run
polymorphic-demo-explainer.pdf	This document — explains what the tool does
demo-script.pdf	Step-by-step demo script with Vision One navigation
README.md	Full technical reference for advanced users

Compatible with: Trend Micro SEPA · Apex One · Vision One XDR · Version 1.0

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